

K2-29b

R-2389

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_250930 · created 2026-08-20T23:28:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460948.9472 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.049
Transit depth (Rp/Rs)^2	2.89 +/- 1.68 [%]
Orbital Inclination (inc)	86.53 +/- 2.41
Airmass coefficient 1 (a1)	5.3 +/- 2.77
Airmass coefficient 2 (a2)	-1.73 +/- 1.71
Scatter in the residuals of the lightcurve fit is	49.47 %
Best Comparison Star	#3 - [246, 82]
Optimal Aperture	1.46
Optimal Annulus	6.25
Transit Duration (day)	0.083 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.049 vs 0.1419 (deviation 0.57σ)
Duration fit vs published	0.083 d vs 0.0925 d (deviation 0.33σ)
Mid-time O–C	-21.1 min
Depth SNR	3.5
Residual scatter	49.47 %

Light curve

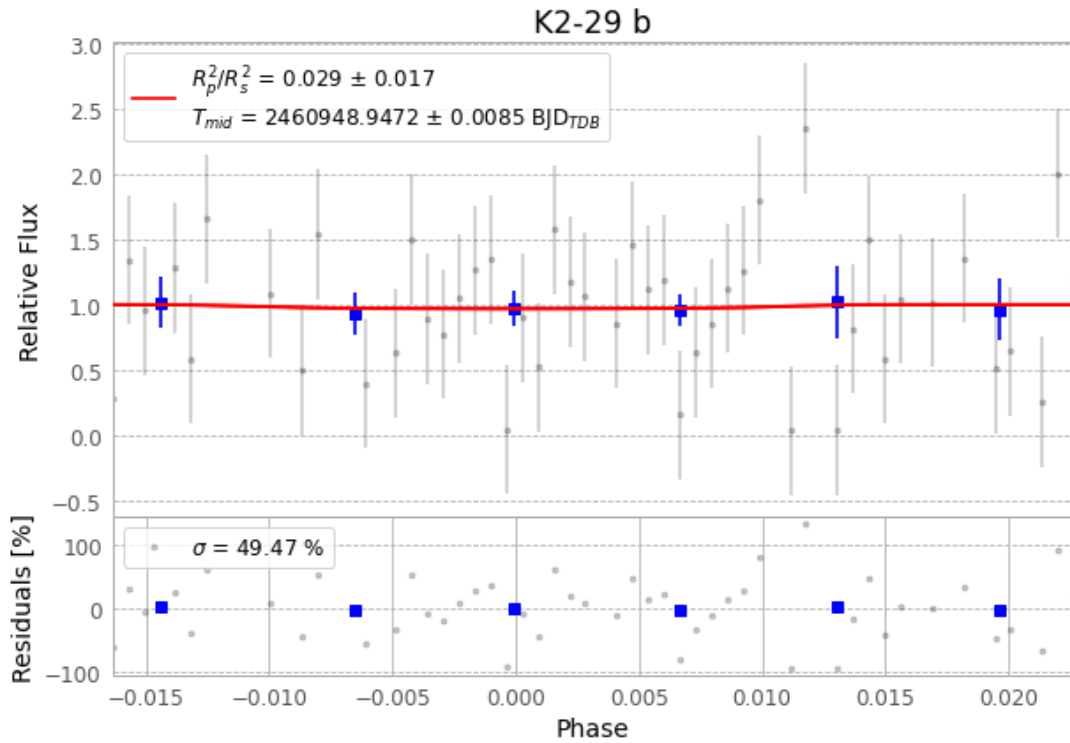


Figure 1 — FinalLightCurve_K2-29 b_2025-09-30.png (SHA-256 099ced985f023b3a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 446], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83e07bfe6c9ae52f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), K2-29250930092116.FITS → K2-29250930122415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-250930.log (98bf4623ef31...), FinalLightCurve_K2-29 b_2025-09-30.png (099ced985f02...), FinalLightCurve_K2-29 b_2025-09-30.csv (b6c13d0a406e...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:28Z.